



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Sunday, 30 August 2026, 7:09 PM**State** Finished**Completed on** Sunday, 30 August 2026, 7:10 PM**Time taken** 59 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two arrays `array_One[]` and `array_Two[]` of same size `N`. We need to first rearrange the arrays such that the sum of the product of pairs (1 element from each) is minimum. That is $\text{SUM}(A[i] * B[i])$ for all `i` is minimum.

For example:

Input	Result
3	28
1	
2	
3	
4	
5	
6	

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int compare_asc(const void *a, const void *b) {
5     long long valueA = *(const long long *)a;
6     long long valueB = *(const long long *)b;
7     if (valueA < valueB) return -1;
8     if (valueA > valueB) return 1;
9     return 0;
10 }
11
12 int compare_desc(const void *a, const void *b) {
13     long long valueA = *(const long long *)a;
14     long long valueB = *(const long long *)b;
15     if (valueA < valueB) return 1;
16     if (valueA > valueB) return -1;
17     return 0;
18 }
19
20 int main() {
21     int n;
22     if (scanf("%d", &n) != 1) {
23         return 0;
24     }
25
26     long long *array_One = (long long *)malloc(n * sizeof(long long));
27     long long *array_Two = (long long *)malloc(n * sizeof(long long));
28     if (array_One == NULL || array_Two == NULL) {
29         return 1;
30     }
31
32     for (int i = 0; i < n; i++) {
33         scanf("%lld", &array_One[i]);
34     }
35     for (int i = 0; i < n; i++) {
36         scanf("%lld", &array_Two[i]);
37     }
38
39     qsort(array_One, n, sizeof(long long), compare_asc);
40     qsort(array_Two, n, sizeof(long long), compare_desc);
41
42     long long min_sum = 0;
43     for (int i = 0; i < n; i++) {
44         min_sum += array_One[i] * array_Two[i];
45     }
46
47     printf("%lld\n", min_sum);
48
49     free(array_One);
50     free(array_Two);
51     return 0;
52 }
```

	Input	Expected	Got	
✓	3 1 2 3 4 5 6	28	28	✓
✓	4 7 5 1 2 1 3 4 1	22	22	✓
✓	5 20 10 30 10 40 8 9 4 3 10	590	590	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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